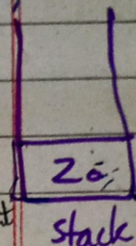
Lecture # 16 & 17

Date: Oct 14, 2025 Day: Tuesday

Push Down Automata (PDA)

↳ for CFL (context free lang)



Finite Automata + Stack ⟹ PDA.

Λ- NFA + stack

7 Tuples: q_0; Q, Σ, δ, $F_{\{\}}$, Γ (gamma), Z_0 (Top element)

$\Gamma = \{Z_0, A, B, \ldots\}$

state alphabet + stack element

$\delta_{PDA} \Rightarrow Q \times (\Sigma \cup \Lambda) \times \Gamma \rightarrow P(Q \times \Gamma^*)$

q_0 (a, b, Λ) (Z_0, A)

$Q = \{q_0, q_1\}$; $\Sigma = \{a, b\}$; $\Gamma = \{Z_0, A\}$

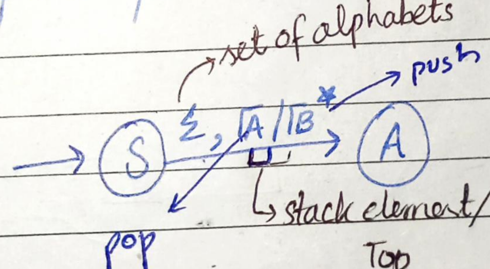
next state + stack element

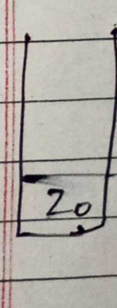
*** Operations for stack:**

push, pop, replace, no-operation, multipush-operation.

Notations:

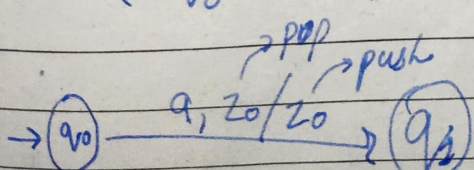
$\Gamma = \{Z_0, A\}$





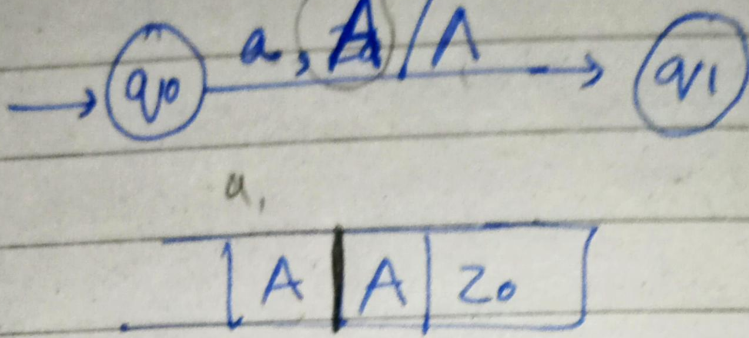
Push ⟹ $\delta(S, \Sigma, Z_0) \rightarrow P(Q, \Gamma^*)$

$\delta(q_0, a, Z_0) \rightarrow q_1,$

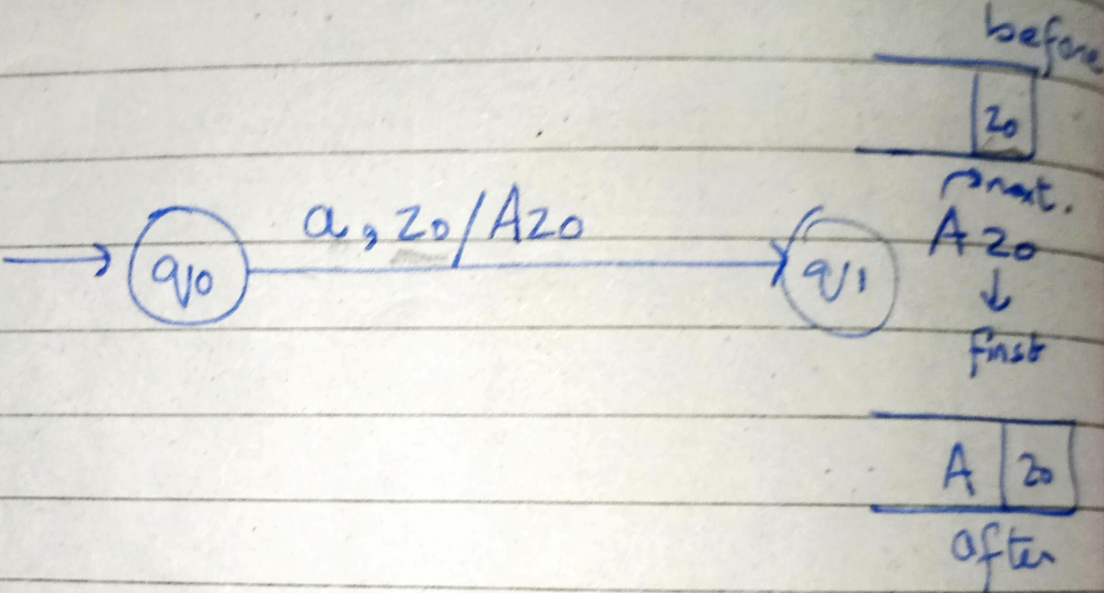


No-Operation

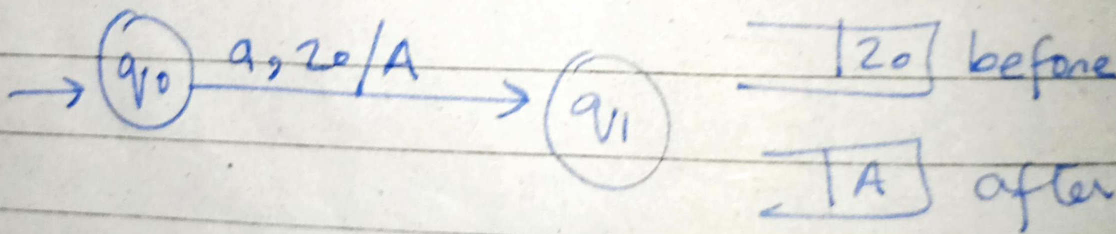
Pop:



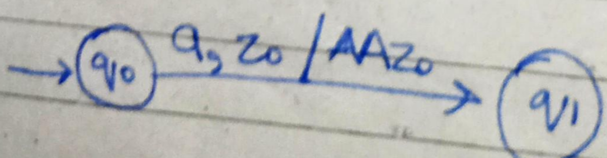
Push:



Replace:



MultiPush:-



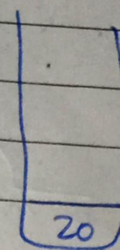
Date: Day:

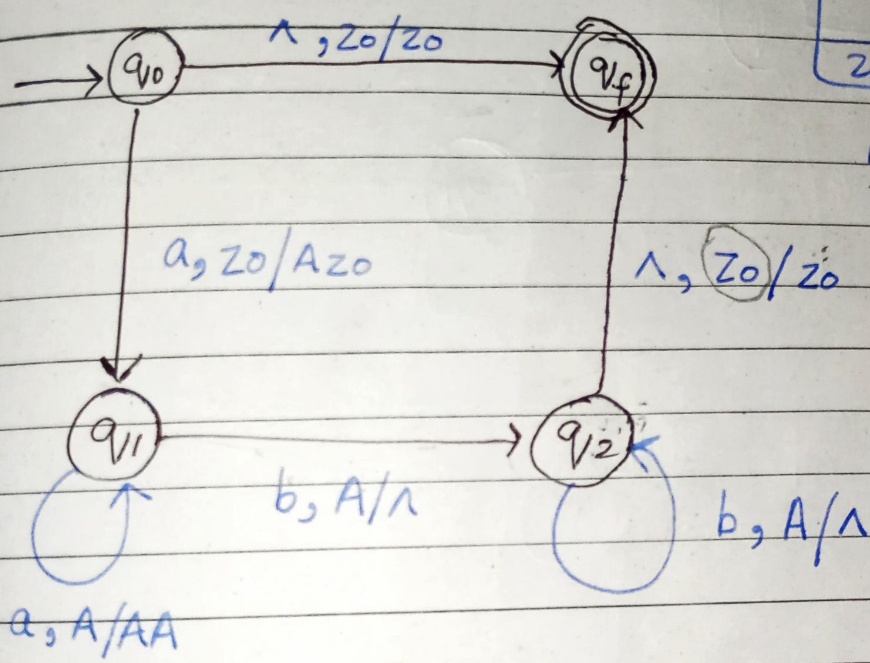
$= \{z_0, A\}$

$Q =$

$\star$ $L = \{a^n b^n \; ; \; n \geq 0\}$

$L = \{\wedge, ab, aabb, aaabbb, \ldots\}$

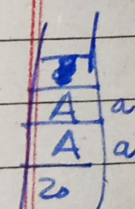


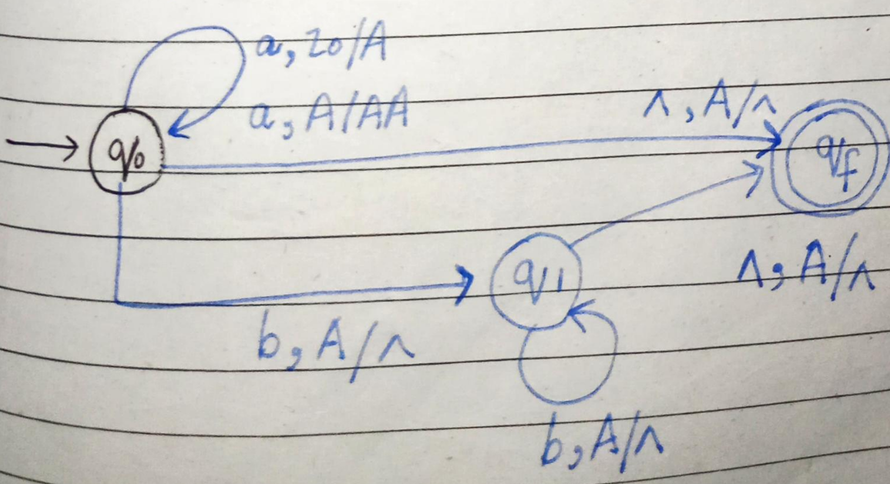


All missing transitions are mapped to dead state.

$\star$ $L = \{a^n b^m \; ; \; n > m \; ; \; m \geq 0\}$

$= \{a, aab, aaab, aaabb, \ldots$ (aa, aaa, aaaa)



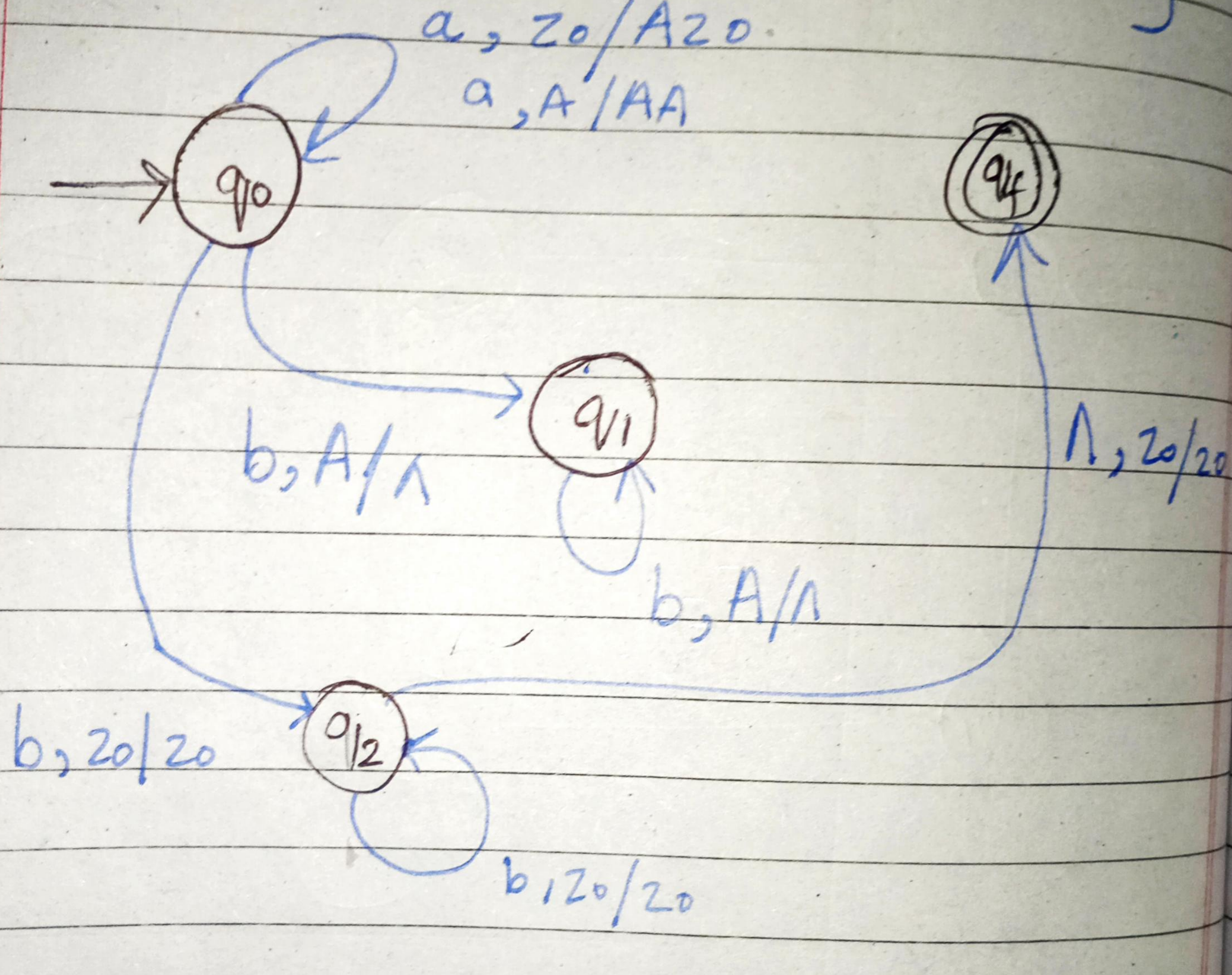


Date:

Day:

$L = \{a^n b^m \; ; \; m > n \; ; \; n \geq 0\}$

$L = \{b, bb, bbb, abb, bbbb, abbb, bbbbb, \ldots\}$



domain

top element

updated stack

Range of the pdA

next state?

Move #	State ①	Input ②	Stack Symbol ③	Stack Status ⑤	Move ④	
1	S	a	Z_0	AZ_0	① (S, AZ_0)	push (A)
2	S	b	A	Z_0	③ (S, Λ)	pop(A)
3	S	b	Z_0	BZ_0	(S, BZ_0)	②
4	S	a	B	Z_0	(S, Λ)	④
5	S	a	Z_0	AZ_0	(S, AZ_0)	①
6	S	a	A	AAZ_0	(S, AA)	⑥
7	S	b	A	AZ_0	(S, Λ)	③
8	S	b	A	AZ_0	(S, Λ)	3
9	S	Λ	Z_0	Z_0	(F, Z_0) Accepted.	⑦

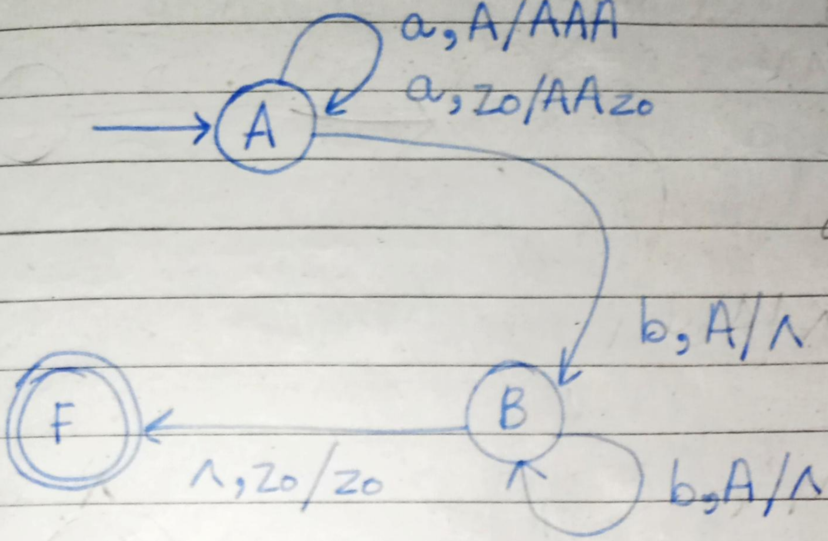
String processing:

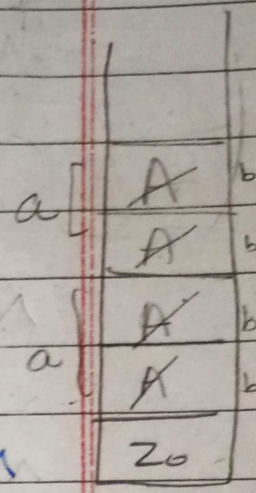
Lecture # 20

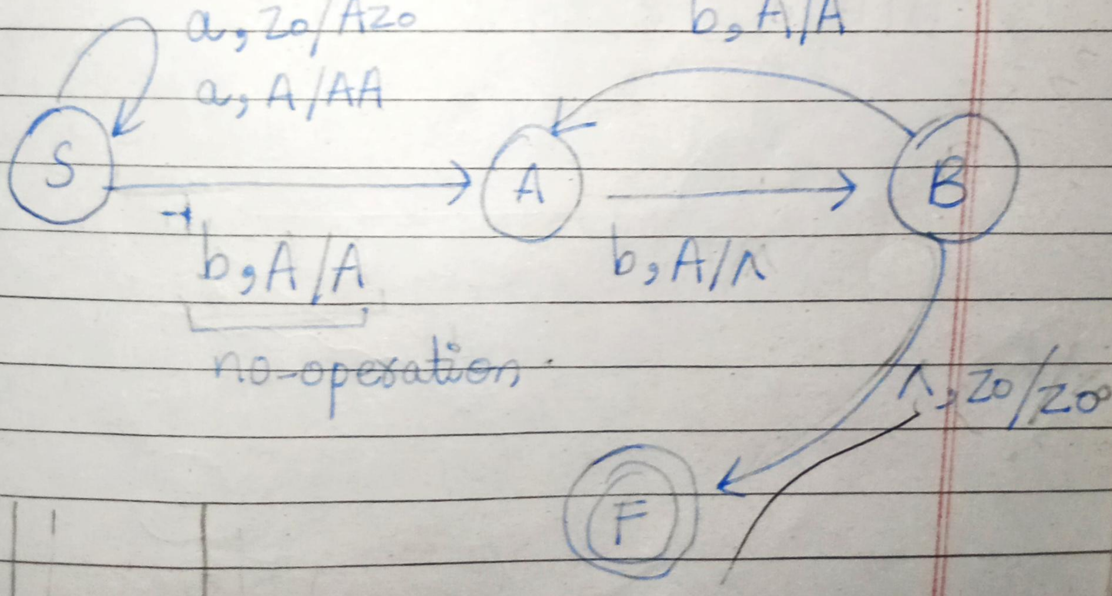
Date: Oct 23, 25 Day: Thursday

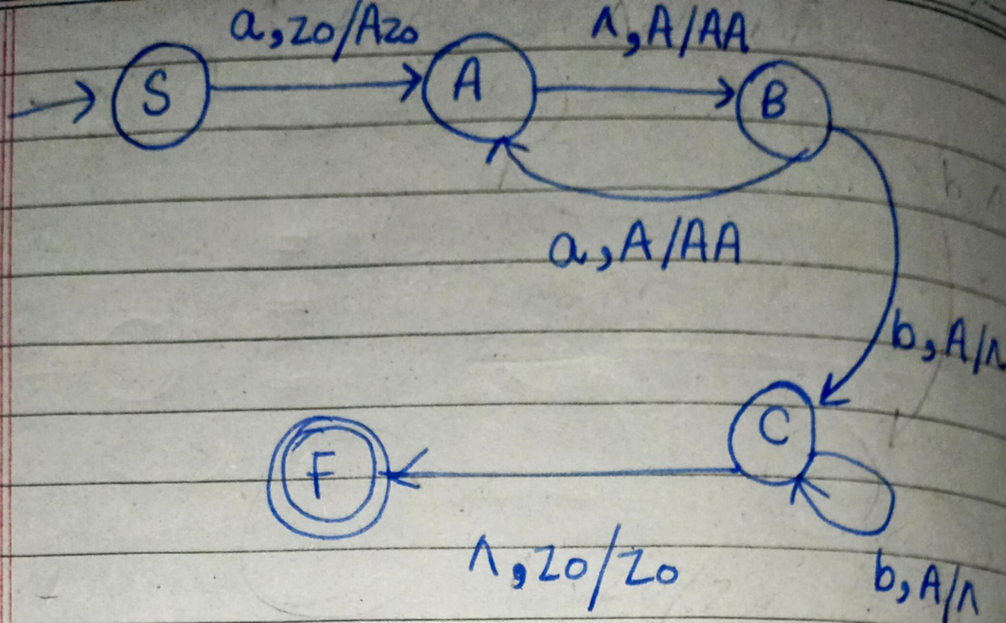
$$L = \{ a^n b^{2n} ; n > 0 \}$$ ~~aa~~bbbb

$L = \{abb, aabbbb, ...$



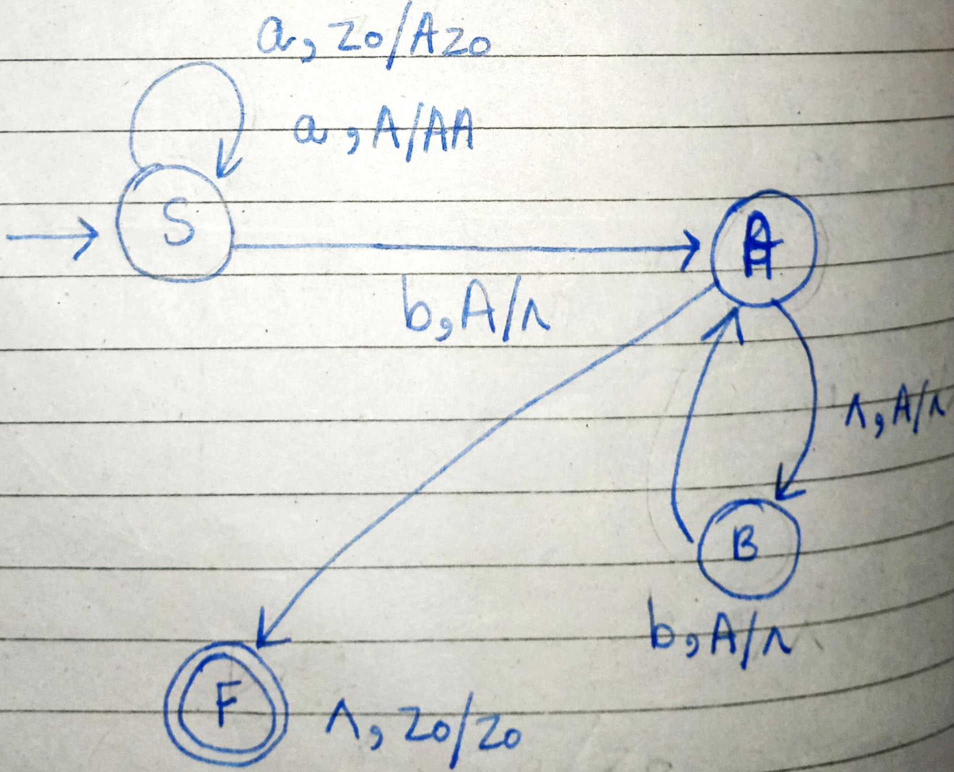






$L = \{a^{2n} b^n \;;\; n > 0\}$

aab, aaaabb, ------



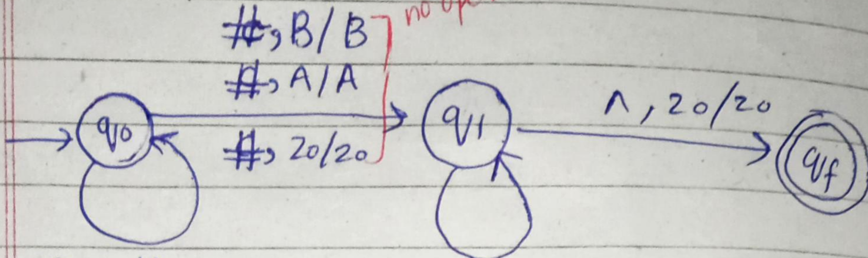
Lecture #21

Date: Oct 29, 25

Day: Thursday

$\Sigma = \{a, b\} + \#$

$L = \{ W \# W^R,\ W \in \{a, b\}^* \}$



1) a, Z0/AZ0
2) a, A/AA
3) b, Z0/BZ0
4) b, B/BB
5) a, B/AB b, A/BA

a, A/^
b, B/^

ab~~ba~~#bba

$L = \{ \#,\ a\#a,\ b\#b,\ aa\#aa,\ bb\#bb,\ ab\#ba,\ ba\#ab, \dots \}$

~~A~~
~~B~~
A

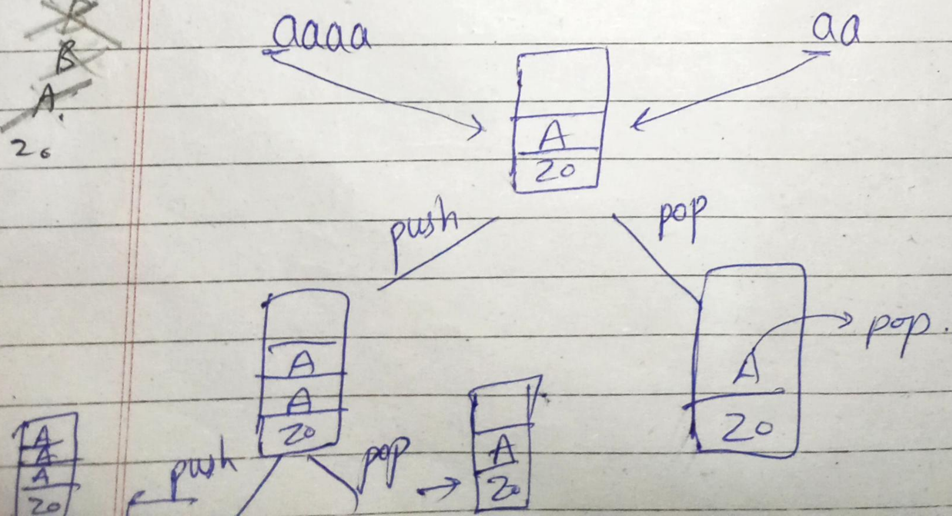
abb # bba

$Q \times \Sigma \times \Gamma \rightarrow P(Q \times \Gamma^*)$

NPDA

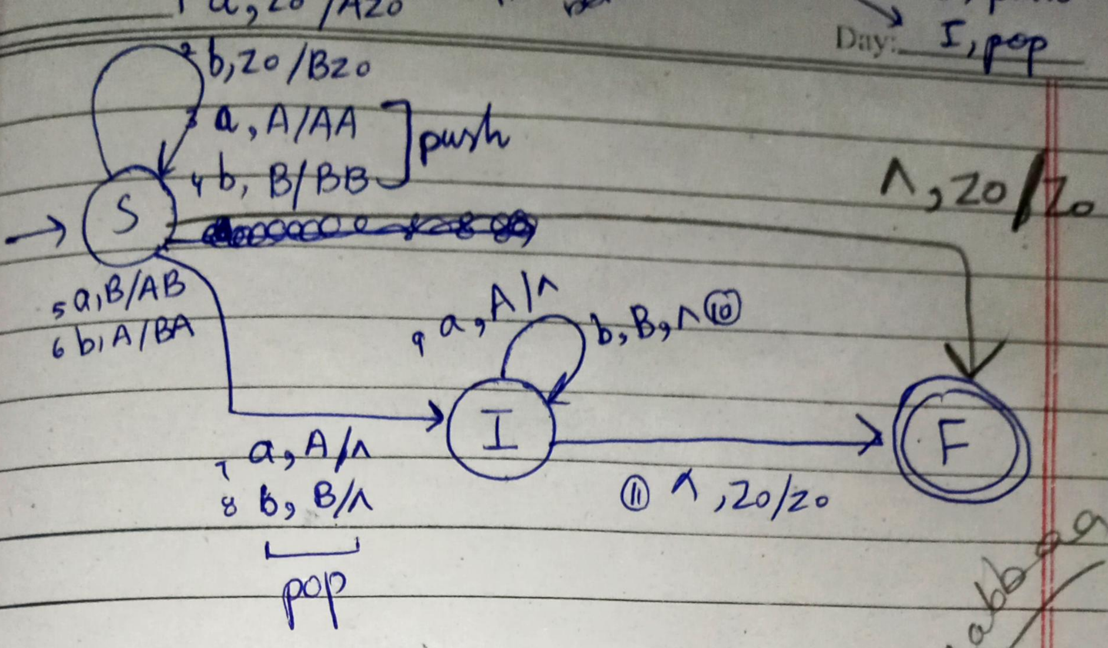
$L = \{ WW^R,\ W \in \{a, b\}^* \}$

copy of stack is made?

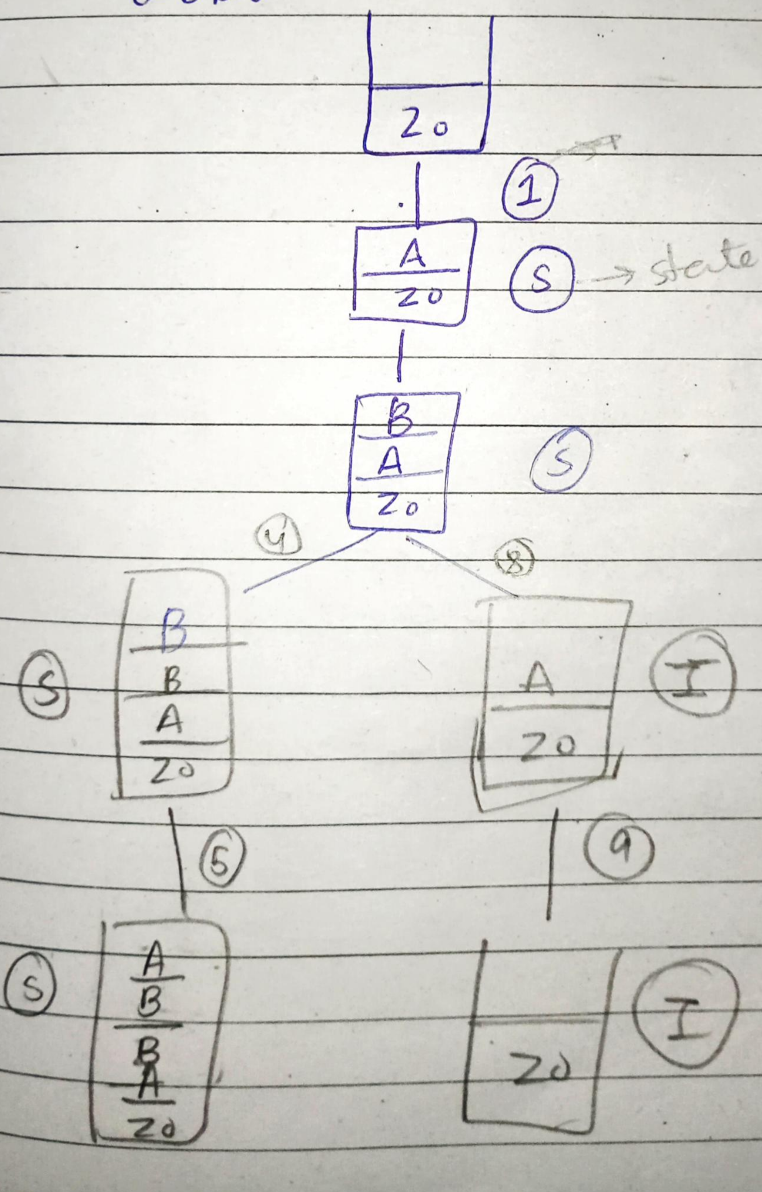


Date:

Day: I, pop



abba

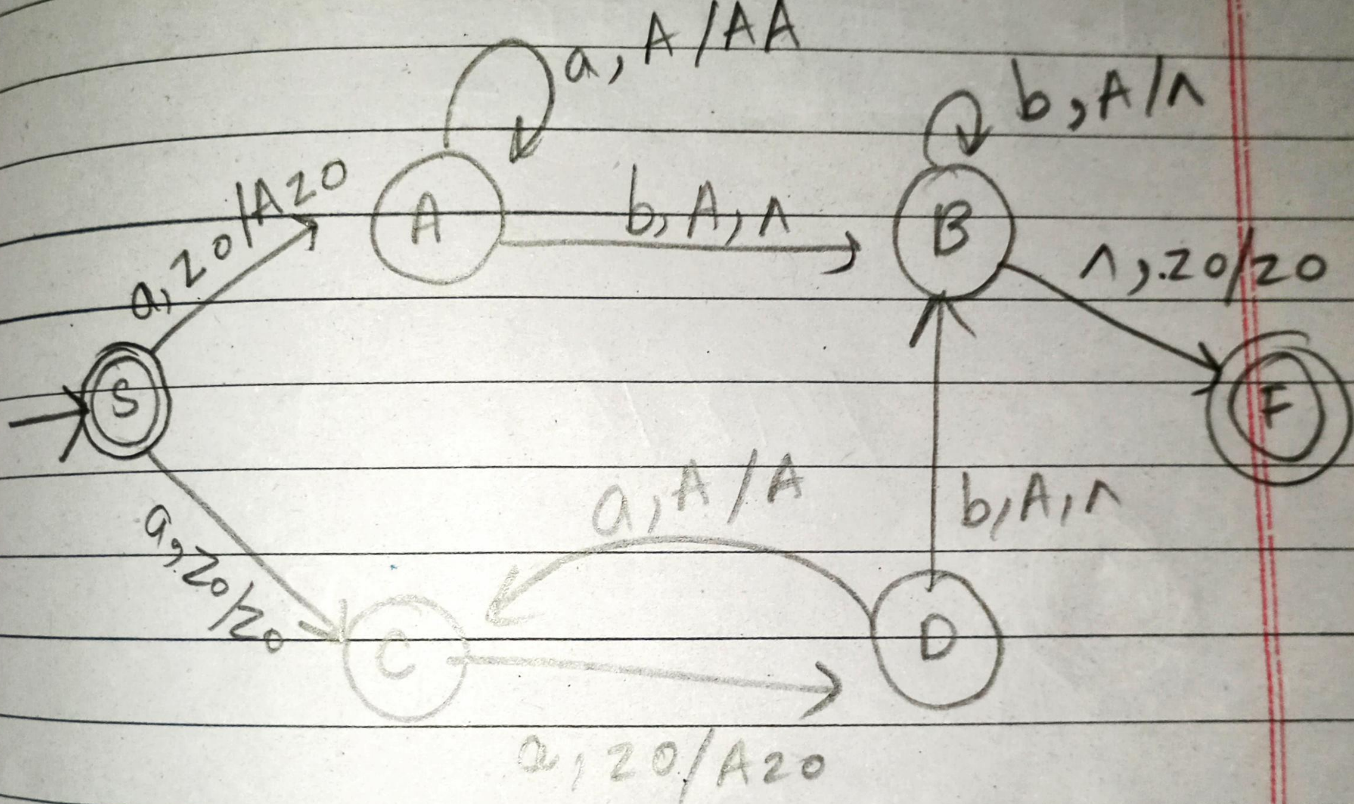


$\Sigma = \{a, b\} + \#$

Date: Day:

$L = \{ a^i b^j \ ; i = j \text{ or } i = 2j \}$

$L = \{ \Lambda, ab, aab, aabb, aaaabb, \ldots \}$



aaaabb free

Date: Day:

$$L = \{ w_i \in \{a, b\}^* \mid x \in w_1 \# w_2 \; ; |w_1| = |w_2| \text{ or } n_a(w_1) = n_b(w_2) \}$$

$$L = \{ \#, b\#, a\#a, a\#a, a\#b, b\#a, b\#b, \}$$